

Research Roadmap

Physical Hallucination Detection in PINN-Generated 2D Navier-Stokes Flow Fields

Undergraduate Research Project Roadmap

1 Project Objective

The goal of this project is to develop a compact and reproducible benchmark for detecting **physical hallucinations** in PINN-generated two-dimensional incompressible Navier-Stokes flow fields.

A **physical hallucination** is defined as a generated flow field that appears visually plausible but violates one or more governing physical constraints, such as momentum conservation, incompressibility, periodic boundary consistency, or expected energy decay.

The project is designed to support a short technical paper of approximately four pages, in the style of an IEEE Letter. Therefore, the scope must remain narrow, focused, and experimentally clean.

2 Central Research Question

The main research question is:

Can a normalized physics-based score detect visually plausible but physically inconsistent perturbations of PINN-generated 2D incompressible Navier-Stokes flow fields?

The expected contribution is not a new PINN architecture and not a better CFD solver. The contribution is a small, interpretable detection framework for identifying physically inconsistent generated flow fields.

3 Scope of the Project

3.1 Included

The project includes:

- 2D incompressible Navier-Stokes equations.
- Taylor-Green Vortex as an analytically controlled flow generator.
- One baseline PINN model.
- Controlled perturbations that create physically invalid but visually plausible fields.
- A Physical Hallucination Score.
- Detection experiments and comparison against residual-only baselines.
- Final figures and tables suitable for a short paper.

3.2 Excluded

The project does not include:

- New PINN architectures.
- Stream-function PINNs.
- Fourier feature PINNs.
- Adaptive collocation.
- Manufactured Solutions.
- 3D Navier-Stokes.
- Turbulent flow claims.
- External CFD datasets.
- Comparison against CFD solvers.

Keeping this scope narrow is essential for completing a publishable short paper.

4 Mathematical Setup

We consider the two-dimensional incompressible Navier-Stokes equations over the periodic domain

$$(x, y) \in [0, 2\pi]^2, \quad t \in [0, T].$$

The velocity field is

$$\mathbf{u}(x, y, t) = \begin{bmatrix} u(x, y, t) \\ v(x, y, t) \end{bmatrix},$$

and the pressure field is $p(x, y, t)$. Assuming unit density, the governing equations are

$$u_x + v_y = 0,$$

$$u_t + uu_x + vu_y = -p_x + \nu(u_{xx} + u_{yy}),$$

$$v_t + uv_x + vv_y = -p_y + \nu(v_{xx} + v_{yy}).$$

The corresponding residuals are defined as

$$R_u = u_t + uu_x + vu_y + p_x - \nu(u_{xx} + u_{yy}),$$

$$R_v = v_t + uv_x + vv_y + p_y - \nu(v_{xx} + v_{yy}),$$

$$R_c = u_x + v_y.$$

A generated flow field is denoted by

$$\hat{s}(x, y, t) = (\hat{u}(x, y, t), \hat{v}(x, y, t), \hat{p}(x, y, t)).$$

A field is classified as physically hallucinated if

$$PHS(\hat{s}) > \tau,$$

where PHS is the Physical Hallucination Score and τ is a threshold selected from valid validation fields.

5 Synthetic Flow Generator

The valid flow fields are generated using the Taylor-Green Vortex:

$$u = U_0 \sin(kx + \phi_x) \cos(ky + \phi_y) e^{-2\nu k^2 t},$$

$$v = -U_0 \cos(kx + \phi_x) \sin(ky + \phi_y) e^{-2\nu k^2 t},$$

$$p = \frac{U_0^2}{4} [\cos(2kx + 2\phi_x) + \cos(2ky + 2\phi_y)] e^{-4\nu k^2 t}.$$

The case parameters are sampled as follows:

$$U_0 \sim \mathcal{U}(0.5, 2.0),$$

$$k \in \{1, 2, 3\},$$

$$Re \sim \text{LogUniform}(20, 300),$$

$$\phi_x, \phi_y \sim \mathcal{U}(0, 2\pi).$$

The viscosity is computed by

$$\nu = \frac{U_0}{Re \cdot k},$$

and the final time is

$$T = \min \left(2.0, \frac{1}{2\nu k^2} \right).$$

6 Baseline PINN

A single baseline PINN is used. The model receives

$$(x, y, t)$$

as input and predicts

$$(\hat{u}, \hat{v}, \hat{p}).$$

The architecture is:

- Fully connected MLP.
- 6 hidden layers.
- 64 neurons per hidden layer.
- tanh activation.
- Float64 precision.

The PINN loss is

$$L = \lambda_{NS}L_{NS} + \lambda_{div}L_{div} + \lambda_{IC}L_{IC} + \lambda_{BC}L_{BC} + \lambda_pL_p.$$

The loss components are:

$$L_{NS} = MSE(R_u) + MSE(R_v),$$

$$L_{div} = MSE(R_c),$$

$$L_{IC} = MSE(\hat{u}(x, y, 0) - u(x, y, 0)) + MSE(\hat{v}(x, y, 0) - v(x, y, 0)),$$

$$L_{BC} = MSE(\hat{s}_{x=0} - \hat{s}_{x=2\pi}) + MSE(\hat{s}_{y=0} - \hat{s}_{y=2\pi}),$$

$$L_p = mean(\hat{p})^2.$$

Use the following loss weights:

$$\lambda_{NS} = 1, \quad \lambda_{div} = 10, \quad \lambda_{IC} = 10, \quad \lambda_{BC} = 10, \quad \lambda_p = 1.$$

7 Physical Hallucination Generation

Hallucinated fields are created by applying controlled perturbations to valid PINN-generated fields.

Use the perturbation strengths

$$\epsilon \in \{0.005, 0.01, 0.02, 0.05, 0.1\}.$$

The perturbations are summarized in Table 1.

Table 1: Perturbation types used to generate physically hallucinated fields.

Type	Perturbed Quantity	Expected Violation
Velocity-divergence	u	Divergence, momentum
Momentum	u, v	Momentum residual
Pressure	p	Pressure-momentum consistency
Temporal mismatch	u, v shifted in time	Temporal and energy consistency
Boundary	u near boundary	Periodic boundary consistency

The five perturbations are:

7.1 Velocity-Divergence Perturbation

$$\tilde{u} = u + \epsilon U_0 \sin(3x + 0.7) \sin(2y),$$

$$\tilde{v} = v, \quad \tilde{p} = p.$$

7.2 Momentum Perturbation

$$\tilde{u} = u + \epsilon U_0 \sin(4x) \cos(3y) \cos(2t),$$

$$\tilde{v} = v + \epsilon U_0 \cos(2x) \sin(5y) \sin(t),$$

$$\tilde{p} = p.$$

7.3 Pressure Perturbation

$$\tilde{u} = u, \quad \tilde{v} = v,$$

$$\tilde{p} = p + \epsilon U_0^2 \cos(5x) \cos(4y).$$

7.4 Temporal Mismatch Perturbation

$$\tilde{u}(x, y, t) = u(x, y, t + \Delta t),$$

$$\tilde{v}(x, y, t) = v(x, y, t + \Delta t),$$

$$\tilde{p}(x, y, t) = p(x, y, t),$$

where

$$\Delta t = \epsilon T.$$

7.5 Boundary Perturbation

Define

$$m(x) = e^{-x^2/\sigma^2} + e^{-(2\pi-x)^2/\sigma^2},$$

with

$$\sigma = 0.2.$$

Then

$$\tilde{u} = u + \epsilon U_0 m(x) \sin(3y),$$

$$\tilde{v} = v, \quad \tilde{p} = p.$$

8 Physical Hallucination Score

For each field, compute four physical violation components.

The momentum violation is

$$S_{mom} = MSE(R_u) + MSE(R_v).$$

The divergence violation is

$$S_{div} = MSE(u_x + v_y).$$

The boundary violation is

$$S_{bc} = MSE(s_{x=0} - s_{x=2\pi}) + MSE(s_{y=0} - s_{y=2\pi}),$$

where

$$s = (u, v, p).$$

The kinetic energy is

$$E(t) = \frac{1}{2} \mathbb{E}_{x,y}[u^2 + v^2].$$

For Taylor-Green Vortex, the expected energy decay is

$$E_{phys}(t) = E(0)e^{-4\nu k^2 t}.$$

The energy violation is

$$S_E = MSE(E(t) - E_{phys}(t)).$$

Each component is normalized using valid validation fields:

$$\bar{S}_j = \frac{S_j}{\text{mean}(S_j^{valid}) + 10^{-12}},$$

where

$$j \in \{mom, div, bc, E\}.$$

The Physical Hallucination Score is

$$PHS = \bar{S}_{mom} + \bar{S}_{div} + \bar{S}_{bc} + \bar{S}_E.$$

The detection threshold is

$$\tau = \text{percentile}_{95}(PHS_{valid}).$$

A field is classified as hallucinated if

$$PHS > \tau.$$

9 Work Plan

The project is organized into six work packages. Each work package has a clear deliverable and acceptance criterion.

Table 2: Project work packages.

WP	Goal	Deliverable
WP1	Mathematical and code foundation	Formulation document, Taylor-Green generator, residual verification test
WP2	Synthetic cases and sampling	30 randomized cases, collocation points, IC points, BC points, evaluation grids
WP3	Baseline PINN	Trained MLP PINN, valid PINN-generated fields, training logs
WP4	Hallucination generation	Five perturbation types, hallucinated field dataset
WP5	Detection score and evaluation	PHS implementation, metrics, baseline comparison, ablation
WP6	Paper assets	Three figures, two tables, short technical report

9.1 WP1: Mathematical and Code Foundation

Goal: Establish the mathematical formulation and verify the analytical generator.

Main actions:

- Implement the Taylor-Green Vortex solution.
- Implement residual computation for R_u , R_v , and R_c .
- Verify that the analytical solution satisfies the governing equations.

Acceptance criteria:

$$MSE(R_u) < 10^{-8},$$

$$MSE(R_v) < 10^{-8},$$

$$MSE(R_c) < 10^{-8}.$$

9.2 WP2: Synthetic Cases and Sampling

Goal: Generate all data needed for PINN training and evaluation.

Main actions:

- Generate 30 randomized Taylor-Green cases.
- Split into 20 training/calibration cases, 5 validation cases, and 5 test cases.
- For each case, generate collocation points, initial condition points, boundary pairs, and a 64×20 evaluation grid.

Acceptance criteria:

- All cases are saved with complete metadata.
- All sampled points are inside the correct domain.
- Evaluation grids are consistent across cases.

9.3 WP3: Baseline PINN

Goal: Train a baseline PINN and generate valid PINN-based flow fields.

Main actions:

- Implement the baseline MLP PINN.
- Implement the PINN loss function.
- Train one PINN per selected case.
- Save predicted u , v , and p fields on the evaluation grid.

Acceptance criteria:

A case is considered usable if

$$RelL2(u, v) < 0.10,$$

$$MSE(R_c) < 10^{-4},$$

$$MSE(R_u) + MSE(R_v) < 10^{-3}.$$

9.4 WP4: Hallucination Generation

Goal: Create visually plausible but physically inconsistent fields.

Main actions:

- Apply the five perturbation types to valid PINN-generated fields.
- Use five perturbation strengths: 0.005, 0.01, 0.02, 0.05, and 0.1.
- Save all hallucinated fields and their labels.

Acceptance criteria:

- Small perturbations remain visually plausible.
- Each perturbation type activates at least one physical violation component.
- The generated dataset includes valid and hallucinated fields.

9.5 WP5: Detection Score and Evaluation

Goal: Evaluate whether the Physical Hallucination Score detects invalid fields.

Main actions:

- Compute S_{mom} , S_{div} , S_{bc} , and S_E .
- Normalize all components using valid validation fields.
- Compute PHS.
- Select the threshold τ from validation fields.
- Evaluate detection on test fields.
- Compare against residual-only baselines.

Baselines:

$$Score_1 = S_{mom},$$

$$Score_2 = S_{mom} + S_{div},$$

$$Score_3 = PHS.$$

Metrics:

- ROC-AUC.
- Precision.
- Recall.
- F1 score.

Acceptance criteria:

$$AUC(PHS) > 0.90.$$

Ideally,

$$AUC(PHS) > AUC(S_{mom} + S_{div}).$$

9.6 WP6: Paper Assets

Goal: Produce final material for a short technical paper.

Required figures:

- **Figure 1:** Valid field, hallucinated field, difference map, and residual heatmap.
- **Figure 2:** Histogram of $\log_{10}(PHS)$ and ROC curve.
- **Figure 3:** Violation signature heatmap showing perturbation types versus score components.

Required tables:

- **Table 1:** Experimental setup.
- **Table 2:** Detection results comparing momentum-only, momentum plus divergence, and full PHS.

Acceptance criteria:

- All figures are publication-ready.
- All tables contain final numerical results.
- Results support the central claim of the project.

10 Experimental Setup Summary

Table 3: Main experimental settings.

Item	Setting
PDE	2D incompressible Navier-Stokes
Valid flow	Taylor-Green Vortex
Domain	$[0, 2\pi]^2$
Cases	30 total
Split	20 train/calibration, 5 validation, 5 test
U_0	$[0.5, 2.0]$
k	$\{1, 2, 3\}$
Re	$[20, 300]$
PINN	MLP, 6 layers, 64 neurons, tanh
Perturbation types	5
ϵ values	$\{0.005, 0.01, 0.02, 0.05, 0.1\}$
Score components	Momentum, divergence, boundary, energy
Baselines	Momentum only, momentum plus divergence

11 Final Success Criteria

The project is successful if all of the following conditions are met:

1. The Taylor-Green analytical generator passes residual verification.
2. The baseline PINN produces visually reasonable flow fields.
3. Small perturbations, especially $\epsilon = 0.01$ or $\epsilon = 0.02$, remain visually plausible.
4. The Physical Hallucination Score increases with perturbation strength.

5. Full PHS outperforms residual-only baselines.
6. The violation signature heatmap shows that different perturbation types activate different physical violation components.
7. The final figures and tables are sufficient for a short technical paper.

12 Risk Management

12.1 Risk 1: The PINN Does Not Converge

Suggested fixes:

- Reduce Re .
- Use only $k = 1$ or $k = 2$.
- Reduce T .
- Increase the number of collocation points.
- Check input and output normalization.

12.2 Risk 2: Perturbations Are Too Obvious Visually

Suggested fixes:

- Reduce ϵ .
- Use smoother perturbation functions.
- Focus on $\epsilon = 0.01$ or $\epsilon = 0.02$.

12.3 Risk 3: PHS Does Not Improve Over Residual-Only Baselines

Suggested fixes:

- Check component normalization.
- Verify boundary and energy score computations.
- Confirm that temporal and boundary perturbations are included.
- Normalize using valid PINN-generated validation fields.

12.4 Risk 4: Valid PINN Fields Already Have High PHS

Suggested fixes:

- Improve PINN training.
- Use lower Re cases.
- Exclude failed PINN cases using a clearly documented rule.
- Report the exclusion rule transparently.

13 Final Deliverables

At the end of the project, the repository should contain:

- Taylor-Green Vortex generator.
- Residual verification tests.
- Randomized case metadata.
- Trained baseline PINN models.
- Valid PINN-generated flow fields.
- Hallucinated flow fields.
- Physical Hallucination Score implementation.
- Detection metrics.
- Baseline comparison.
- Three final figures.
- Two final tables.
- Short technical report summarizing the results.